

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2025

Course: B.Tech	Branch: Mechanical Engineering	Semester: VI
Subject Code & Name: BTME603C and I C Engines		
Max Marks: 60	Date:16/01/2026	Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The function of a carburetor in an SI engine is to a. Cool the engine b. Supply lubricating oil c. Prepare air–fuel mixture d. Increase compression ratio	CO1	1
2	The following is a S.I. engine a. Diesel engine b. Petrol engine c. Gas engine d. None	CO2	1
3	Indicated power of single cylinder 2-stroke engine is equal to a. $pLAN$ b. $pLAN/2$ c. $4pLAN$ d. $2pLAN$	CO3	1
4	Compression ratio of a diesel engine is usually in the range of ? a. 14-22 b. 10-12 c. 6-8 d. 25-30	CO2	1
5	Which fuel has the highest octane rating? a. Diesel engine b. Petrol engine c. Gas engine d. LPG	CO4	1
6	Which alternative fuel is commonly used in fuel cell vehicles? a. Hydrogen b. Ethanol c. LPG d. Methanol	CO3	1
7	Detonation in SI engines is caused due to: a. Higher carbon emissions b. Auto-ignition of end gas c. High engine speed d. Lower emission of pollutants	CO3	1
8	Which component is essential in an electric vehicle (EV)? a. Carburetor b. Exhaust pipe c. Fuel injector d. Lithium-ion battery	CO4	1
	What is <i>regenerative braking</i> in electric vehicles?		

9	a. A way to boost engine power	b. A feature to reduce tire wear	c. A method of applying hydraulic brakes	d. A system that charges the battery during braking	CO3	1
10	Which country is the largest EV market as of recent years?				CO5	1
	a. USA	b. India	c. Japan	d. China		
11	Which pollution norm is currently followed in India for new vehicles (as of 2025)?				CO5	1
	a. BS-III	b. BS-IV	c. Euro-V	d. BS-VI		
12	Which pollutant is <i>not</i> typically regulated under BS emission norms?				CO4	1
	a. Carbon monoxide (CO)	b. Nitrogen oxides (NOx)	c. Carbon dioxide (CO ₂)	d. Helium (He)		
Q. 2	Solve the following.					12
A)	Differentiate between SI engine and CI engine				CO3	6
B)	Draw valve timing diagrams for 4-S Diesel (Theoretical and actual)				CO2	6
Q.3	Solve the following.					12
A)	In an ideal Otto cycle, the air at the beginning of isentropic compression is 3 bar and 25 degrees centigrade. The compression ratio is 8. If the heat added during the constant volume process is 240 KJ/Kg Determine 1. Max. Temp of the cycle 2. Efficiency 3. Work done				CO5	6
B)	What is knocking and detonation in IC engines				CO4	6
Q. 4	Solve Any Two of the following.					12
A)	Explain the magneto ignition system of engine with a neat sketch https://www.batuonline.com				CO3	6
B)	Explain the Stages of Combustion for 4 Stroke Petrol Engine				CO4	6
C)	Explain the Supercharging and Turbo Charging with a neat sketch				CO3	6
Q.5	Solve Any Two of the following.					12
A)	List any four advantages of electric vehicles				CO3	6
B)	Why lubrication is required in the IC Engine.				CO5	6

C)	Explain the operating principles, construction, and working of Fuel Cell Technology	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	State the different exhaust emissions of an IC engine.	CO4	6
B)	Write the main components of electric vehicles	CO5	6
C)	Describe the Series and Parallel Electric Vehicle with a neat sketch	CO6	6
	*** End ***		